OBJECT-ORIENTED PROGRAMMING
Winter semester acad. year 2019-2020
1st set of exercises

Exercise 1: The present exercise aims to train you in construction - destruction functions and class composition.

Implement in C++ a representation of a discussion forum, e.g. "Object-Oriented Programming", as it is described below. For reasons of simplicity, assume that there are no subforums.

A forum consists of its "title" and a number of threads.

Each thread consists of a "subject", its "creator", its "creation date" and a number of posts.

Each post consists of a "numeric identifier" (id, unique for each post), its "title", its "creator", its "creation date" and a text.

When a post is created, the values of its elements are assigned and printed on the screen. When it is destroyed, it prints on the screen the message, `"Post number:` ⟨*Post_number*⟩ `is about to be destroyed"`, where ⟨*Post_number*⟩ is the numeric identifier of the post.

When a thread is created, its subject, its creator and its creation date are assigned. Also, it prints on the screen the message `"Thread with subject:` ⟨*Subject*⟩ `has just been created!"`, where ⟨*Subject*⟩ is the subject of the thread. When it is destroyed, it prints on the screen the message, `"Thread with subject:` ⟨*Subject*⟩ `is about to be destroyed"`, where ⟨*Subject*⟩ is the subject of the thread.

When a forum is created, its title is assigned and it prints the message `"Forum with title:` ⟨*Title*⟩ `has just been created!"`, where ⟨*Title*⟩ is the title of the forum. When it is destroyed, it prints on the screen the message, `"Forum with title:` ⟨*Title*⟩ `is about to be destroyed"`, where ⟨*Title*⟩ is the subject of the forum.

As regards the behavior of the above, the following hold. The forum should provide the ability to be printed, by printing the titles of its threads. Also, it should provide the ability to select a thread, by specifying the title and then the thread should be printed. Finally, the forum should provide the ability to select a post, by giving its number and then this post should be printed.

A thread is printed, by printing its subject, its creator, its creation date and its posts.

A post is printed, by printing its number, its title, its creator, its creation date and its text.

Implement the above, through the appropriate classes, defining the data members that are needed and the member functions that implement the above behavior. Also implement a function `main` in which:

1. A forum with title "Object-Oriented Programming" is created
2. The forum is printed
3. The thread with subject "Rules and useful information" is printed
4. Some posts are selected (randomly), and printed.

Note 1 **:** The texts for the posts should be created in a simple way, e.g. have an array of expressions and randomly select some of them.

Note 2 **:** For the representation of the date, consider a structure.

Note 3 **:** Wherever you need strings, use the type `string` from the Standard library of C++ (the necessary declarations are in the header file `string`).

Note 4 **:** The "creators" should be represented by a string that corresponds to their name in the forum.

Exercise 2: [1] The behavior of the above forum should be enhanced with one more print function (`print_sorted`) which prints all its posts, sorted in ascending order by creator. The posts of the same creator, among themselves, need not be sorted in any way.

For the sorting use a binary search tree structure, with nodes that consist of the creators accompanied by a list of their posts. To create this structure, for each post, if there is no node that corresponds to its creator, a new node should be inserted in the appropriate position. If it already exists, the post is simply added to the list.

The tree should be initialized taking a thread as input, inserting all its posts. Once it has been created, it should be possible to insert a new post into the tree as described above. When it is destroyed, it should be ensured that all the memory allocated for its creation is freed.

Also, it should be possible to print the tree, by printing the content of the nodes, as the type of their parts indicates. The printing should be done in the order of visiting the nodes "depth first and from left to right" inorder traversal.

Finally, the ability to enhance the tree with the contents of another tree should be offered. For each node of the second tree, if for its creator there is a node in the first tree, its posts should be added to the list of posts of the first. Otherwise, a new node should be created in the first tree that contains the content of the node. You can see the result of this process in Figure 1.

So, taking into account the above structure, the print function `print_sorted` for a forum:

1. creates a tree from one thread of the forum,
2. enhances it with the trees that result from the other threads of the forum and, finally,
3. prints the tree.

Also implement a function `main` in which:

1. A forum with title "Object-Oriented Programming" is created
2. The forum is printed using the above print function.

[1] I thank Orestis Polychroniou for suggesting the tree structure

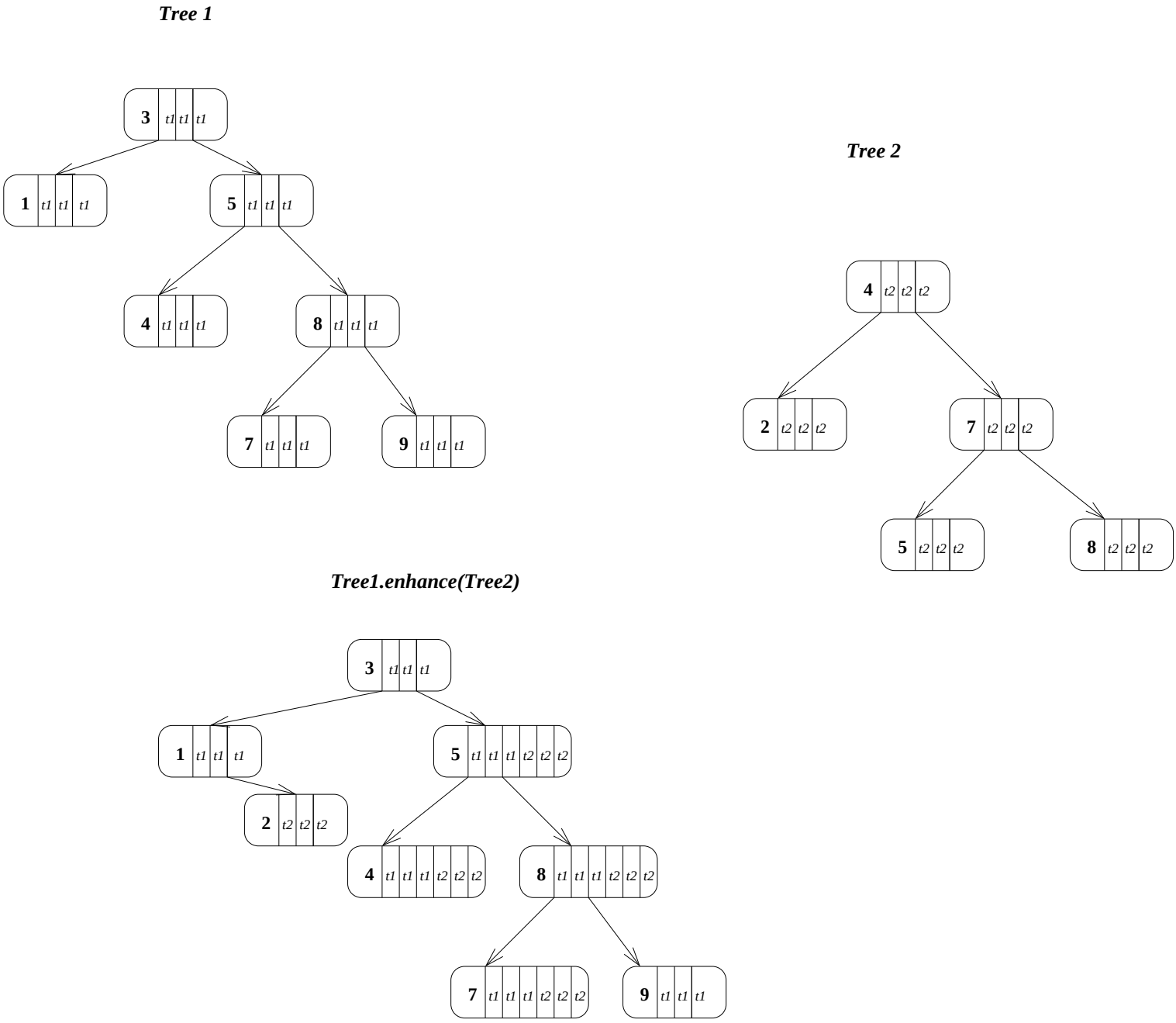


Figure 1: Result of enhancing tree Tree1 with the contents of tree Tree2